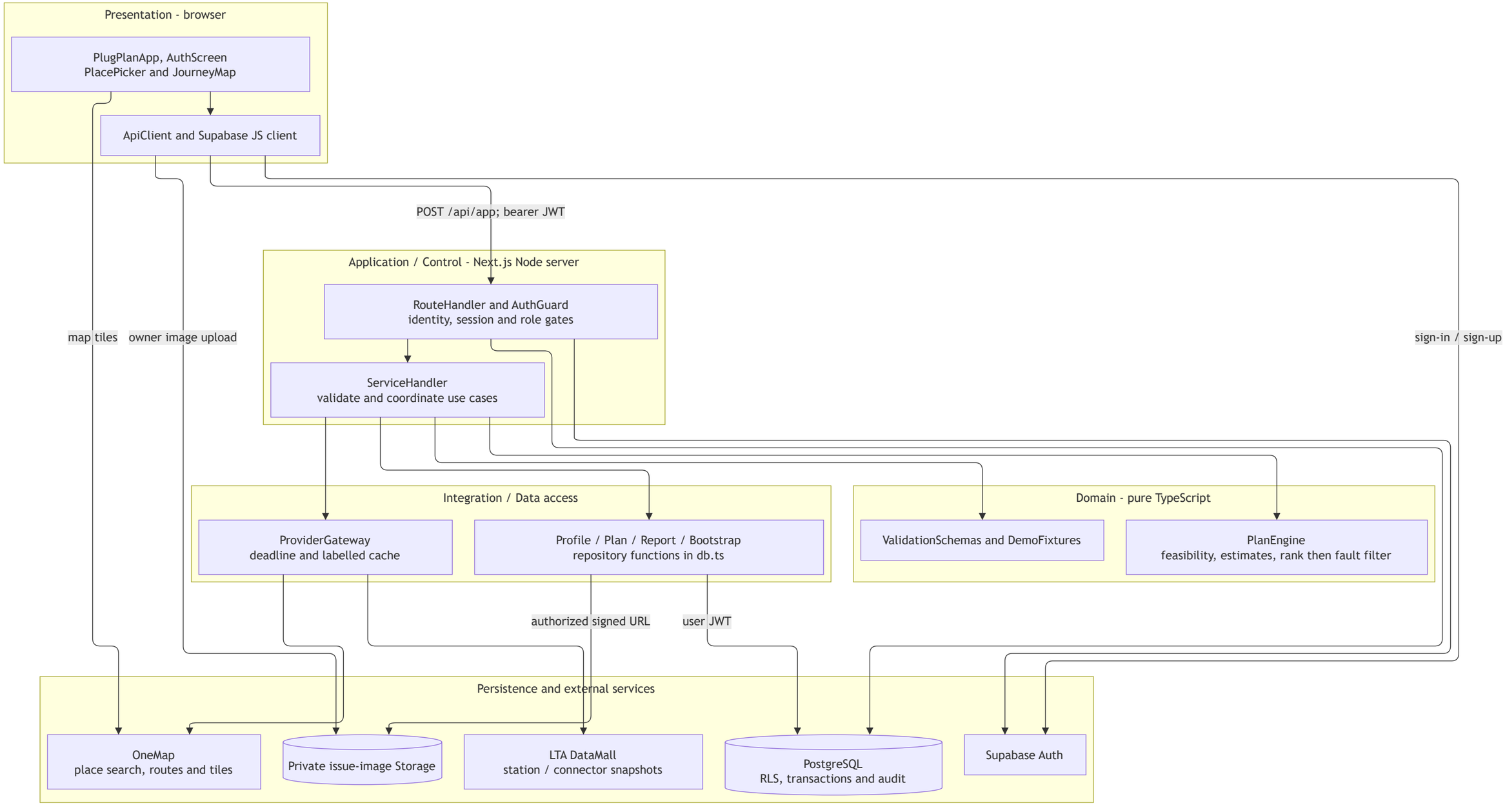
PlugPlan SG — System Architecture

1. Architectural style

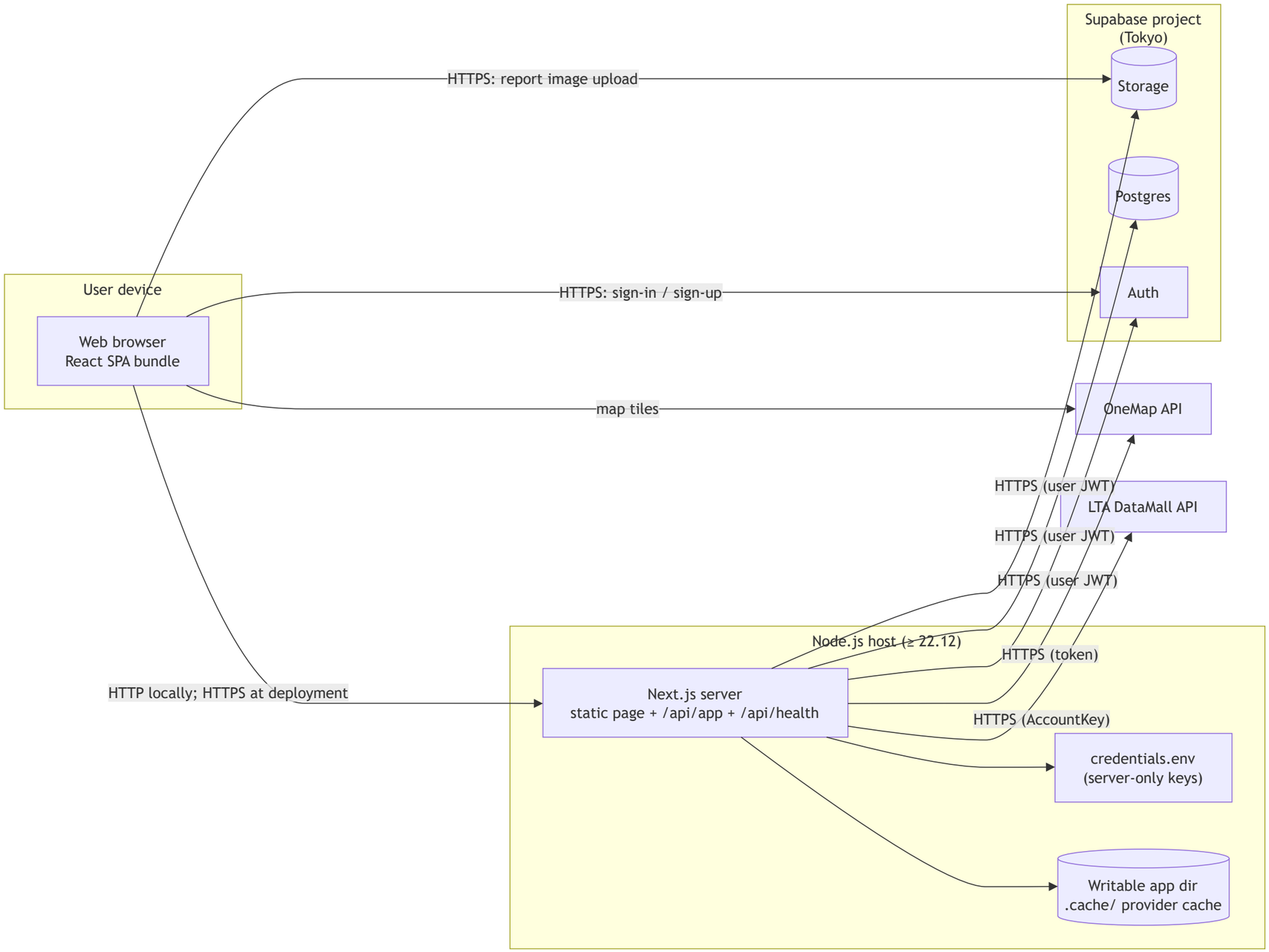
PlugPlan SG uses a **layered client–server architecture** with five responsibility areas. The server coordinates domain and integration work, while the browser also calls Supabase Auth and Storage directly and requests OneMap map tiles; this implementation therefore has explicit cross-layer service paths.

Layer	Responsibility	Code
Presentation	Browser UI: views, forms, map, client-side state	`src/components/plugplan-app.tsx`, `journey-map.tsx`, `src/app/page.tsx`
Application / Control	Single API entry point, authentication, role gates, input validation, use-case coordination	`src/app/api/app/route.ts`, `src/lib/server/service.ts`, `supabase.ts`, `errors.ts`
Domain	Pure charging-plan calculation (feasibility, energy, cost, ranking), shared schemas, demo fixtures	`src/lib/domain/engine.ts`, `validation.ts`, `fixtures.ts`, `src/lib/types.ts`
Integration / Data access	Repositories over Supabase tables and RPCs; gateway to external government APIs with timeout and cache	`src/lib/server/db.ts`, `src/lib/server/providers.ts`
Persistence / External services	Postgres with RLS and stored procedures, Auth, private Storage; LTA DataMall; OneMap	`supabase/migrations/*.sql`; third-party HTTPS APIs

2. Layered architecture diagram

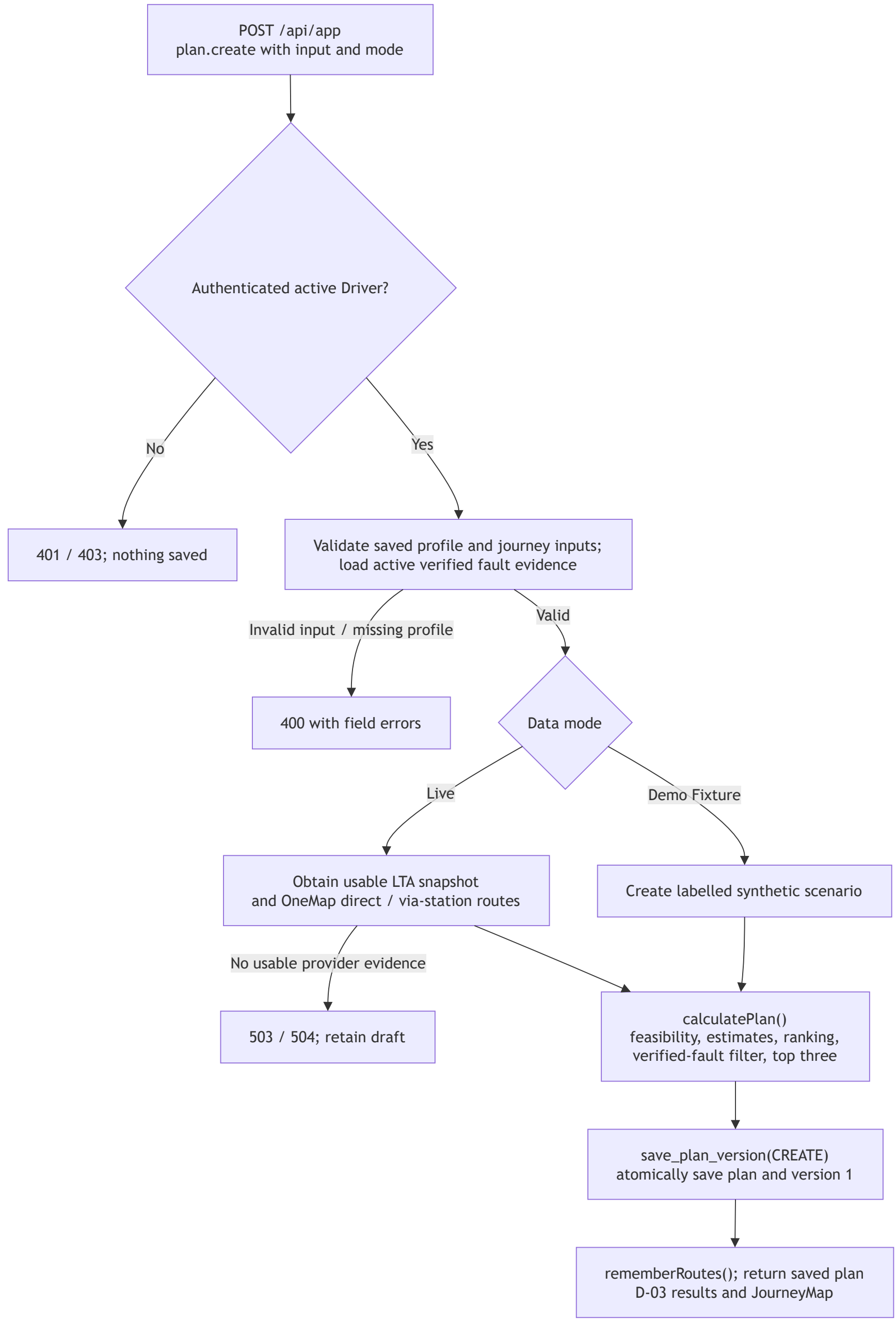


3. Deployment view



- LTA and OneMap credentials exist only on the server. The browser receives only the Supabase URL and the public (publishable) key.
- The server never holds a Supabase service-role key. Every database call carries the end user's JWT, so database policies always apply.
- The current delivery is a local demonstration on `127.0.0.1:3000`. A production deployment needs HTTPS at the host or reverse proxy.

4. Request flow: create a charging plan (UC-03)



For UC-03 plan creation, an LTA cache fallback does not skip OneMap: the route lookup must succeed before the plan is saved. Existing-plan refresh has a separate recovery path that may reuse a recent saved route after a provider failure and labels the result with a warning.